

# FoodLoop - System Architecture

---

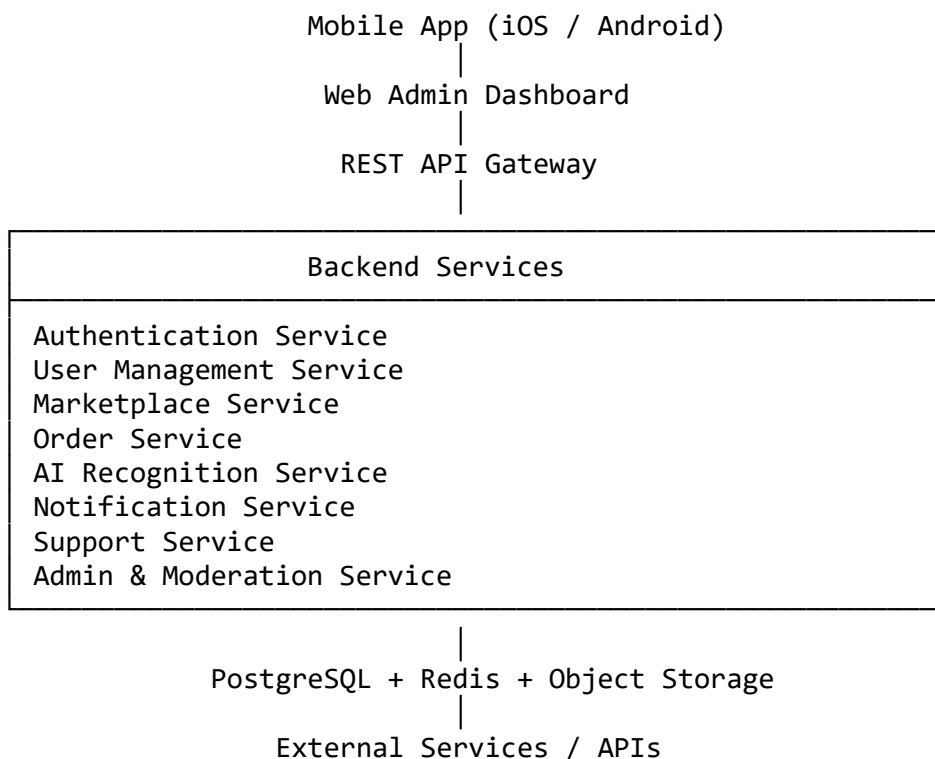
## 1. Overview

FoodLoop is a marketplace that connects consumers with nearby businesses selling surplus food at discounted prices. The platform aims to reduce food waste while providing affordable food options through AI-assisted product recognition, personalized recommendations, delivery management, and trust & safety mechanisms.

The system is designed using a modular architecture where each domain is isolated into its own service or module. This allows the platform to scale independently as the user base grows.

---

## 2. High-Level Architecture



## 3. Architecture Principles

The platform follows several design principles:

- Modular service architecture
  - API-first development
  - Stateless backend services
  - Secure authentication using JWT
  - Event-driven notifications
  - Scalable cloud deployment
  - AI services isolated from business logic
  - Clear separation between consumer, merchant, courier, and admin roles
- 

## 4. Core Services

### 4.1 Authentication Service

Responsible for:

- User registration
- Login
- Password reset
- Email verification
- JWT access tokens
- Refresh tokens
- Role-based authorization

Roles:

- Consumer
  - Merchant
  - Courier
  - Administrator
- 

### 4.2 User Management Service

Responsible for:

- User profiles
- Addresses

- Preferences
  - Languages
  - Saved payment methods
  - Notification settings
- 

## 4.3 Marketplace Service

The main service of the application.

Responsibilities:

- Product listings
  - Deal management
  - Categories
  - Search
  - Filtering
  - Nearby businesses
  - Product availability
  - Favorites
- 

## 4.4 Order Service

Handles the complete purchasing workflow.

Responsibilities:

- Shopping cart
- Checkout
- Payment processing
- Order creation
- Order history
- Refund requests
- Order status tracking

Typical order statuses:

- Pending
- Confirmed
- Preparing
- Ready
- Out for Delivery

- Delivered
  - Cancelled
- 

## 4.5 AI Recognition Service

Provides AI-assisted product identification.

Responsibilities:

- Image upload
- OCR processing
- Product classification
- Expiration date extraction
- Confidence scoring
- Manual review when confidence is low

This service is primarily used by merchants when creating product listings.

---

## 4.6 Notification Service

Centralized notification management.

Notification channels:

- Push notifications
- Email
- In-app notifications

Examples:

- New order
  - Delivery updates
  - Order ready
  - Promotions
  - Support replies
  - Password reset
  - Verification emails
-

## 4.7 Support Service

Handles customer support.

Responsibilities:

- Ticket creation
  - Customer conversations
  - File attachments
  - Issue tracking
  - Escalation to administrators
- 

## 4.10 Admin Service

Provides platform management.

Responsibilities:

- User management
  - Merchant verification
  - Report moderation
  - Product moderation
  - Analytics
  - Audit logs
  - System monitoring
- 

## 5. External Integrations

The platform may integrate with external providers.

| Integration                | Purpose                                |
|----------------------------|----------------------------------------|
| Maps API                   | Store locations, geocoding, navigation |
| Payment Gateway            | Online payments                        |
| Email Service              | Verification & password reset          |
| Push Notification Provider | Mobile notifications                   |
| Object Storage             | Product images                         |
| AI Models                  | OCR & image recognition                |

---

## 6. Data Storage

### PostgreSQL

Stores relational business data including:

- Users
  - Products
  - Orders
  - Stores
  - Reviews
  - Tickets
- 

### Redis

Used for:

- Session caching
  - Frequently accessed data
  - Rate limiting
  - Temporary verification codes
  - Queue support
- 

### Object Storage

Stores:

- Product images
  - Store logos
  - User profile pictures
  - Review photos
  - Support attachments
- 

## 7. Security

The platform implements several security measures:

- JWT authentication
- Refresh tokens

- Password hashing
  - HTTPS-only communication
  - Role-based access control
  - Input validation
  - Rate limiting
  - Audit logging
  - Secure file uploads
- 

## 8. User Roles

### Consumer

Can:

- Browse deals
  - Purchase products
  - Track orders
  - Leave reviews
  - Contact support
- 

### Merchant

Can:

- Manage store
  - Create listings
  - Manage inventory
  - Process orders
  - View analytics
- 

### Administrator

Can:

- Manage users
- Verify merchants
- Handle disputes
- Moderate listings

- View platform analytics
  - Monitor system activity
- 

## 9. Typical User Flow

